Question 1.

Write a program to carry out addition, division, subtraction, and multiplication of three numbers. The program

should also compute the average of all the numbers.

The program should give the following output assuming the numbers are 40, 20, and 30;

The sum: 40 + 20 + 30 = 90

The product: 40 * 20 * 30 = 2400

The difference: 30 – 20 = 10

The quotient: 40 / 20 = 2

The average of 40, 20, and 30 = (40 + 20 + 30)/3 = 30

Ensure that you are able to explain the working of every statement in your program code.

Question 2.

a. Write a C program that accepts a set of five floating point numbers entered from the keyboard by the user,

computes their sum, average, product, and prints the following output on the screen:

You entered the following numbers: <print all the numbers entered by the user>

The sum of the numbers is: <print the sum of the five numbers>

The average of the numbers is: <print the average of the five numbers>

The product of the numbers is: <print the product of the five numbers>

All numbers should be up to 2 decimal points.

b. Write a C program that reads three numbers, finds the smallest and the largest of the three numbers, and prints

the value of the smallest and the largest number on the screen.

Sample output assuming the three numbers are 10, 20, and 30.

You Entered 10, 20, 30

The smallest number is: 10

The largest number is: 30

Question 3.

a) Write a C program to find the sum, average, and the largest of any three numbers (marks) input by the user.

The program should also assign a grade to the average of the three marks based on the following guidelines:

Above or equals 70 marks: A

Between 60 and 69 marks: B

Between 50 and 59 marks: C

Between 40 and 49 marks: D

Less than 40: E

The program should not grade an input of more than 100 or less than 0 marks. The message "Invalid Marks"

should be displayed to the user in case of any such entries.

Sample output assuming the user entered 56, 75, and 63.

Marks Entered 56, 75, and 63.

Highest Score: 75

Lowest Score: 56

Average Score: (56 + 75 + 63)/3 = 194/3 = 64.67

Grade for Average Score: B

b) Write a C program that calculates weekly wages for hourly employees. Regular hours 0-40 are paid at KES

1,000/hour. Overtime (> 40 hours per week) is paid at 150% of the regularly hourly payments.

Sample output assuming an employee worked for 48 hours in a week:

Total Number of Hours: 48 hours

Regular Hours: 40 Hours

Overtime Hours: 8 hours

Regular Pay: 40 x 1000 = 40000

Overtime Pay: (8 x 1000 x 150)/100 = 12000

Total Pay: 40000 + 12000 = 52000

Note: the number of hours worked in a week should entered by the user from the keyboard.

Question 4.

a) Write a C program using for loops to print all the numbers between 0 and 10, inclusive. The program

should produce output as shown below;

num = 0

num = 1

num = 2

num = 3

num = 4

num = 5

num = 6

num = 7

num = 8

num = 9

num = 10

b) Write a C program using for loops to print and sum all the odd numbers between n and m, inclusive.

Note that n and m are the lower and upper ranges respectively and should be entered by the user. The

program should produce output as shown below; (assume n=0 and m =10).

Number Sum

num = 1 0 + 1 = 1

num = 3 1 + 3 = 4

num = 5 4 + 5 = 9

num = 7 9 + 7 = 16

num = 9 16 + 9 = 25

Sum of odd numbers between 0 and 10 is equal to 25

c) Using for loops write a C program that implements the following computation:

x=n * n-1 * n-2 *……*1 where n is an integer number entered by the user.

For example, if the number entered is 6 then it computes:

6*5*4*3*2*1

and produces the answer 720

Your program should produce the following output:

Sample output:

You Entered: 6

6 * 5 * 4 * 3 * 2 * 1 = 720

d) Using for loops write a C program to generate the following sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, . . .

. .and so forth, where the term $xn = xn-1 + xn-2$. and the first two terms are 0 and 1.

Sample output:

Enter the term: 9

The sequence up to 9th term is: 0, 1, 1, 2, 3, 5, 8, 13, 21

Question 5.

a. Write a C program (strictly using a do while loop) to generate the following sequence 0, 1, 1, 2, 3, 5, 8,

13, 21, 34, ...and so forth, where the term $xn = xn-1 + xn-2$.

Sample output:

Enter the term: 9

The sequence up to 9th term is: 0, 1, 1, 2, 3, 5, 8, 13, 21

b. Write a C program (strictly using a while loop) to compute the total number of even integers between n

and m (inclusive), where n is less than m. The range (n to m) should be entered from the keyboard by

the user.

Sample output: assume n=2 and m=16

The total number of even integers between 2 and 16 is: 8

c. Write a C program (strictly using a for loop) to print the following pattern:

0 1 2 3 4 5

0 1 2 3 4

0 1 2 3

0 1 2

0 1

0

0 1

0 1 2

0 1 2 3

0 1 2 3 4

0 1 2 3 4 5

d. Write a C program (strictly using a do while loop) to compute the sum all the numbers that are multiples

of 5 between n and m (inclusive), where n is less than m. The range (n to m) should be entered from

the keyboard by the user.

Sample output: assume n=10 and m=30

The sum of multiples of 5 between 10 and 30 is:

10 + 15+ 20 + 25 + 30 = 100

e. Write a C program to print the following pattern:

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* *
* * *
* * * *
* * * * *
* * * * *
* * * *
* * *
* *
*
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Question 6.

a. Write a C program that receives two sets of numbers (vect_one and vect_two) as input and adds them

up to produce a third set of numbers (vect_three). Each set contains 4 numbers. For example, if vect_one

= (3,4,5,6) and vect_two = (6,5,4,3) then vect_three = (3,4,5,6) + (6,5,4,3) = (9,9,9,9). Note that all the

inputs are variables that should be input by the user from the keyboard. Your output should look like the sample output shown below;

Input:

Enter the elements of vect_one: 3 4 5 6

Enter the elements of vect_two: 6 5 4 3

Output:

Vector One: (3 4 5 6)

Vector Two: (6 5 4 3)

Vector Three = Vector One + Vector Two =: (3 4 5 6) + (6 5 4 3) = (9 9 9 9)

b. Write a C program that receives two sets of numbers (vect_one and vect_two) as input and combines

them to produce a third set of numbers (vect_three). vect_one and vect_two contain n numbers, where

n is a value entered from the keyboard by the user. For example, if n=4 and vect_one = (3,4,5,6) and

vect_two = (6,5,4,3) then vect_three = (3,5,4,6,6,5,4,3). Note that all the inputs are variables that should

be input by the user from the keyboard. Your output should look like the sample output shown below:

Input:

Enter the dimension: 4

Enter the elements of vector1: 3 4 5 6

Enter the elements of vector2: 6 5 4 3

Output:

Vector1: (3 4 5 6)

Vector2: (6 5 4 3)

Vector 3 = (3 5 4 6 6 5 4 3)

c. Using arrays, write a C program that takes in a set of five numbers entered by the user, compares the numbers, and produces the smallest of the numbers as an output. Your output should look like the

sample output shown below;

Input:

Enter the five numbers: 3 6 1 8 2

Output:

You entered the following five numbers: 3 6 1 8 2

The smallest of 3 6 1 8 and 2 is 1.

d. Write a C program to print the following pattern:

Question 7.

a. Write a C program that takes in a nxm matrix_one and a nxm matrix_two, sums the two matrices and produces

a nxm matrix_three. The dimensions of the two matrices should be keyed in by the user.

Example;

Assuming n=2 and m=3.

Matrix One;

2	3	4
5	6	7

Matrix Two;

3	4	5
6	7	8

Matrix Three;

5	7	9
11	13	15

b. Write a C++ program that takes a n x m matrix_one and produces a (n+1) x (m+1) matrix_two that consists of

all the elements of the of the first matrix plus their row and column totals. The size and elements of the matrix

must be entered by the user from the keyboard.

Example;

Assuming n=3 and m=3.

Matrix One;

2	3	4
5	6	7
1	8	9

Matrix Two;

2	3	4	9
5	6	7	18
1	8	9	18
8	17	20	45

Question 8.

a. Write a C++ program to compute the circumference and area of a circle. The program should ask the user to

enter a double number corresponding to the radius of the circle, compute the circumference and area, and display

the area and circumference of the circle. Assuming the lengths of the sides of a square are equivalent to twice the

radius of the circle, include additional code in your program to compute the area and perimeter of the square and

display the area and perimeter of the square.

b. Repeat number (1) above using functions. A part form the main() function you should have additional functions

such as f_circum_circle(), f_area_circle(), f_perimeter_square(), and f_area_square().

c. Using arrays, write a C program that accepts any user defined number of inputs (n) and finds the smallest of the

numbers, finds the largest number, sorts the numbers in ascending order, finds the sum of the numbers, and

computes their average.

Example:

Assuming the input is: 7,9,4,1,8,6,2

Output:

The Input List: 7 9 4 1 8 6 2

The Smallest Number: 1

The Largest Number: 9

The Sum: 37

The Average: 5.29

The Sorted List: 1 2 4 6 7 8 9

d. Repeat number (3) above using functions. A part from the main() function you should have additional functions such as find_smallest(), find_largest(), comp_sum(), comp_average(), and sort_ascending().

Question 9.

a.Using arrays, write a C++ program that accepts the following array elements as input,

3,6,7,9,2,4,5,7,9,1,2,4,3,5,7,8,9. The program should allow the user to search the array and produce an output as

highlighted below;

Example 1:

Enter the Search Key: 9

Output:

Number 9 Exists in the List

Number 9 Occurs 3 times

It Appears at Array Indices: 3,8,16

Example 2:

Enter the Search Key: 10

Output:

Number 10 Does not Exist in the List

Number 10 Occurs 0 times

Note: A part from the main() function you should have additional functions such as linear_search() etc.

b. Using arrays, write a C++ program that accepts any user defined number of inputs (n) and sorts the numbers in

ascending order using insertion sort.

Example:

Assuming the input is: 7,9,4,1,8,6,2

Output:

The Input List: 7 9 4 1 8 6 2

Insertion Sort Steps;

Sorted List: 7

Sorted List: 7 9

Sorted List: 4 7 9

Sorted List: 1 4 7 9

Sorted List: 1 4 7 8 9

Sorted List: 1 4 6 7 8 9

Sorted List: 1 2 4 7 8 9

The Sorted List: 1 2 4 6 7 8 9

Note: A part from the main() function you should have additional functions such as sort_ascending() etc.

Question 10.

Read and make notes (in your eee102 exercise book) on the following topics/sections of C++ programming;

a) Basic Syntax

b) Comments

c) Data Types

d) Variable Types

e) Variable Scope

f) Constants/Literals

g) Modifier Types

h) Storage Classes

i) Operators

j) Loop Types

k) Decision Making

l) Functions

m) Arrays